

Practical 10

Aim: Implement Depth First Search (DFS) algorithm for a graph using adjacency matrix representation in C++.

Source-code:

```
#include <iostream>
using namespace std;
int n;
int adj[10][10];
int visited[10];
// DFS function
void DFS(int v) {
    cout << v << " ";
    visited[v] = 1;
    for (int i = 0; i < n; i++) {
        if (adj[v][i] == 1 && visited[i] == 0) {
            DFS(i);
        }
    }
}
int main() {
    int start;
    cout << "Enter number of vertices: ";
    cin >> n;
    cout << "Enter adjacency matrix:\n";
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            cin >> adj[i][j];
        }
    }
    for (int i = 0; i < n; i++) {
        visited[i] = 0;
    }
    cout << "Enter starting vertex: ";
    cin >> start;
    cout << "DFS Traversal: ";
    DFS(start);
    return 0;
}
```

Output:

Enter number of vertices: 4

Enter adjacency matrix:

0 1 1 0

1 0 0 1

1 0 0 1

0 1 1 0

Enter starting vertex: 0

DFS Traversal: 0 1 3 2